

Comparision of Segmentation Techniques for Acute Lymphoblastic Leukemia in Leukemia Cancer

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Abstract—Leukemia is a white blood cancer which is commonly seen in the bone marrow and lymphatic system where the production of abnormal cells which are immature blood cells (blast cells) will dominate the normal cell, this may lead the patient to lose proper functioning of the immune system and may not carry their normal functions, Leukemia is classified according to its speed of progression into Acute and Chronic. Acute is fast-growing and needs aggressive, timely treatment. Lymphocytic leukemia affects lymphoid cell that forms lymphatic tissue which makes the immune system. Acute lymphoblastic leukemia is most commonly seen in young children under 20 and also in adults. Segmentation of ALL cell nuclei with help pathologist in a proper diagnosis of ALL cancer. In this paper, a comparative study of Image processing based edge detection and segmentation techniques with different filters were carried out and the results obtained by each technique have been compared with evaluation metrics like PSNR, MAE and achieved MAE 110.62 and 66 and PSNR 33.2.15 and 10.65 respectively

Index Terms—Acute lymphoblastic leukemia, Acute, Chronic, Segmentation.

I. INTRODUCTION

Leukemia is a blood cancer for which there is no cure and also no proper evidence for its cause there are several factors that may lead the person to get affected by leukemia like it may cause through genetics, radiation etc. cancer can be treated using traditional methods like chemotherapy, targeted therapy, bone marrow transplantation etc, which are a highly economical and time-consuming process. to improve the quality of life for the patient who is undergoing treatment using image processing techniques, segmentation of abnormal cell will be a challenging task for leukemia cancer Segmentation of cell nuclei will help computer-aided diagnosis systems (CADs) to accurately detect cancer cells. the experiment was carried out on image segmentation techniques like Region-based, foreground graph-cut based and cluster-based segmentation techniques and Edge detection techniques like Sobel, Prewitt, Laplacian and Canny were compared and thresholding techniques like Otsu, Global thresholding, adaptive mean and adaptive gaussian was compared. the dataset used for this experiment was ALL IDB2 which was a publicly available dataset that consists of 260 samples with 130 healthy and 130 unhealthy of size 257 X 257. The figure below is the input original image collected from ALL IDB2.

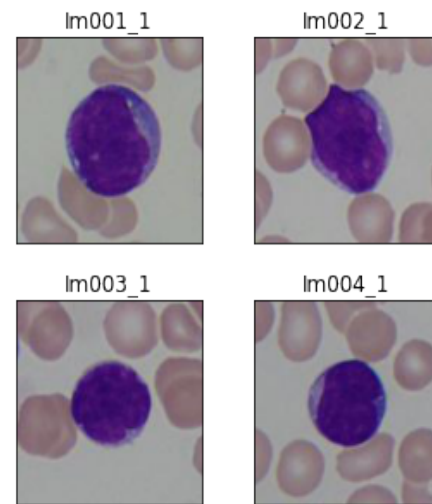


Fig. 1. Image from ALL_IDB2

II. RELATED WORK

S. Mohapatra et al., [1] have proposed a two-stage colour segmentation strategy to separate leukocytes and other blood components. in the second stage feature descriptors like nucleus shape, texture and novel Hausdorff dimension and contour signature were used to detect leukemia and for the classification of lymphocytic cell nucleus SVM was used. Ranjitha P et al., [2] experiments were carried out for detection of cancer stages and classification of lymphocytes, for which preprocessing of microscopic images collected from the Kaggle dataset that contains noise at the central shadow of the image was converted to colour space to removed unwanted noise and to restored wanted noise, Histogram equalizer that improves the contrast and image intensity for better visualization as needed and k-means clustering-based segmentation for segregating. and achieved better performance Nilkanth Mukund Deshpande et al., [3] have proposed an effective and efficient hybrid segmentation technique initially image preprocessing and segmentation were carried out using Multi-level thresholding along with Otsu and

Kapur methods. and to optimize the performance of segmentation Learning enthusiasm-based teaching-learning-based optimization (LebTLBO) algorithm was proposed. the proposed algorithm was compared with the existing benchmark method using performance metrics like PSNR and similarity index. segmentation using LebTLBO showed significant improvement.

F. Soni et al, [4] have done preprocessing by converting the original image to CMYK colour system and L*a*b, where CMYK represents cyan, magenta, yellow and black which are been subtracted from white colour and L*a*b is the commonly used system for image representation, segmentation of nucleus was carried out by hybrid K-means with CMYK and L*a*b and for classification SVM was used, precision, recall, accuracy was measured which results in the performance of hybrid segmentation gives better result compared to k-means

Toh et al, [5] proposed a hybrid technique with colour based thresholding on RGB images of ALL for colour information and a median filter was applied to segment the abnormal cells from the unwanted region like the background.

III. METHODOLOGY

The suggested stages for comparing segmentation techniques are depicted in the block diagram below. Here, preparation operations like scaling and edge detection are carried out on photos obtained from the ALL_IDB dataset. The input photos from ALL IDB2 were first preprocessed by resizing the image to 256 X 256 and the input image is converted to greyscale, and then the appropriate Edge detection and segmentation techniques were performed. Results were executed in Google Colab using Python.

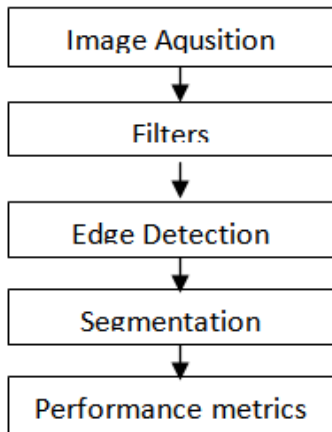


Fig. 2. Steps in the segmentation process as proposed

A. Filters:

Averaging: Averaging is used to smoothen an image by reducing the intensity value of neighbouring pixels, it is done

by moving pixel by pixel and replacing values of neighbour pixels by average value..

$$g[m] = \alpha f[m] + (1 - \alpha)g[m - 1] \quad (1)$$

In this equation, the current output, the prior output, and the current input are denoted as $g[m]$, $g[m-1]$, and $f[m]$, respectively. The α parameter ranges from 0 to 1. $\alpha=1$ If there is no filtering and the output is just equal to the input.

Gaussian: Gaussian is applied to blur the region of images, it can be applied on an odd symmetric kernel that is convolved over each pixel in the image and focus is on a region of interest. Gaussian filter is an approximation of Gaussian function..

$$Gauss[i]_x = \sum_{y \in S} Gauss_{\sigma}(\|x - y\|)i_y \quad (2)$$

$Gauss[i]_x$ is the outcome at pixel x in this case, and the RHS is just a total of all the pixels y that have had the Gaussian function applied to them. The intensity for pixel y is i_y .

Bilateral: Bilateral is non-linear filtering that helps in smoothing the data as well as preserving edges by reducing noise, this filter replaces pixel intensity values with a weighted average. this is used to preserve sharp edges

$$g_x(z) = G^t(x - g(z)) \quad (3)$$

where $g_x(z)=G^t(xg(z))$ and $g_x(z)$ is the "picture." Now, $b(y,g(y))$ gives the value of the bilateral filter at location y . Therefore, for each value of $z=f(y)$ in the image, we should calculate $b(y,z)$ in order to perform the bilateral filter without error.

Median: Median is non-linear filtering that uses the mean value of input pixel under a window and gives output, this is a widely used filter in digital image processing to remove unwanted noise from images.

$$B(p, q) = \text{median}\{I(p, q)\} \quad (4)$$

$B(p, q)$ is the resultant value for pixels p and q for image I .

B. Edge Detection:

Edge detection can be considered as preprocessing step for segmentation; in edge detection the boundaries of region of interest are identified. This boundaries can be recognized by detecting discontinuities in gray level image .edges are set of connected pixels which can form boundaries. Edges can be horizontal, vertical and diagonal.

Edge Detection operators are classified in to Gaussian and Gradient, the operators that are computed using first order derivatives are called gradient like Robert, Sobel and Prewitt

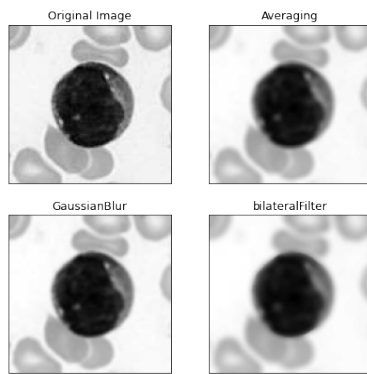


Fig. 3. Resultant images of filters

operators and the operators that are computed using second order derivatives are Laplacian and canny operator.

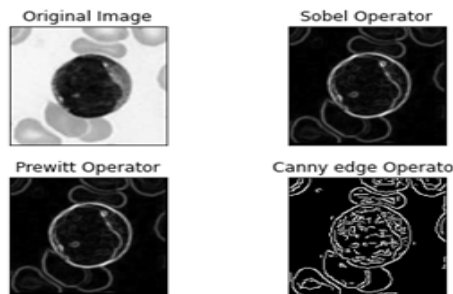


Fig. 4. Edge detection techniques

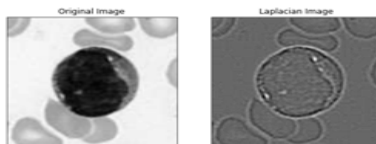


Fig. 5. Laplacian

The Advantages, Limitations and applications of this operator are discussed in table.1

C. Image Thresholding

This is the simplest form of segmentation where the grayscale or colour image is converted to binary form; this is done to separate the objects in the foreground from the background. Normally thresholding can be applied to the entire image, but in some cases, thresholding is needed at some region in this case we can go for local thresholding where the region of interest is at some particular place in the entire image. And Global thresholding is applied if we want the features from the entire image.

The thresholding value should be chosen by the user itself which may be difficult sometimes and may not be appropriate. For this automatic thresholding exists which will choose the

Table 1 Edge Detection

Edge detection	Advantages	Limitations	Applied for Cell
Sobel, Prewitt, Laplacian, Canny	Effective detection of boundaries of images, most of irrelevant data is removed. Canny is useful for extracting structured information from various objects and is easily adaptable to various environments	Noise present in images will slow the performance. Sobel does not produce smooth and thin edges when compared to canny. Laplacian and Prewitt are sensitive to noise. Canny is time consuming and shows bias for edges.	Edge detection can be used for images where edges are uneven & spread through. And having considerable contrast between objects. Sobel and Prewitt can detect edges with orientations. Laplacian can be used to find fine detail for image and restore it. Canny can be applied for images where edges are uneven.

thresholding value by itself based on the image which has to be separated from the background. There are many types of thresholding few of them are automatic which most commonly used thresholding is Otsu.

The Advantages, Limitations and applications of a few popular thresholding techniques are discussed in table 2.

Table 2 Thresholding Technique

Thresholding Technique	Advantages	Limitations	Applied for Cell
Otsu	Requires less space for storage and processing is fast. Many of applications still use this as basic segmentation.	Image should be free from noise and captured in good illumination conditions.	Can be applied for cell which has only two binary classes to segment and with homogeneous intensity.

D. Segmentation

Segmentation is the technique in which the digital image is divided into different segments, these segments are the regions which help in further processing and analyzing the images by which complexity of analysis reduces. There exists different type of segmentation techniques based on their clustering-based, region-based, object-based, seed-based, colour-based, histogram-based and many more. Here k-means which is clustered based segmentation and watershed which is region-based segmentation were used.

- K value is chosen at random for each option, with k=2 being the most effective.

- Step2: According to the k value two-centred pixels are selected at random.
- Step3: Using the Euclidian Distance formula the distance is calculated from each pixel to the centre pixel and the pixels which are having the shortest distance from the centre pixel are assigned to that pixel, this is done with every pixel in the image and k clusters are formed.

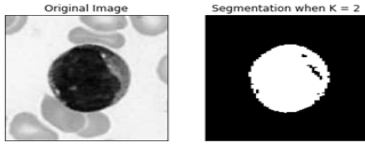


Fig. 6. input gray scale image with segmentation k=2

Marker-based Watershed Segmentation is the region-based segmentation technique where image morphology is utilized and a seed point has to be chosen for each region for segmentation. This will exactly separate the background from the foreground, here for input image binary Otsu thresholding is applied then morphological operators closing 2X2 and dilation was considered and for which distance transform is calculated with Euclidian distance and applied thresholding where unknown regions are marked as boundaries.

experiments were carried out by considering edge detection techniques like Sobel, Prewitt, Laplacian and Canny. as the result Prewitt and Sobel's operator was giving a better result which can be visualized in the figure below

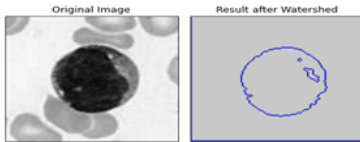


Fig. 7. Result of Marker based watershed

IV. PERFORMANCE METRICS

The most commonly used evaluation metrics in image processing for quantifying images are MSE, MAE, RMSE and PSNR, from which MAE and PSNR are chosen where:

Mean Absolute Error(MAE): MAE is a measure of the average magnitude of errors, that is the absolute value of the difference between actual and predicted divided by total observations

$$MeanAbsoluteError = \frac{1}{m} \sum_{j=0}^m |g_j - z_j| = \frac{1}{m} \sum_{j=1}^m |e_j| \quad (5)$$

Peak Signal to Noise ratio(PSNR): MSE is the cumulative the squared error between the predicted to the original image, PSNR is the measure of peak error, to calculate PSNR,

MSE should be calculated and higher the PSNR better the performance

$$PeakSignaltoNoiseratio = 10 \log_{10} \left[\frac{\max_j^2}{mse} \right] \quad (6)$$

V. RESULTS AND DISCUSSION

the table below shows the calculated value of MAE and PSNR by which we can say that laplacian and watershed are working better for ALL IDB2 images and also can be visualized in a box plot..

Table 3 MAE and PSNR for Filters

Original Image	Averaging	Gaussian	Bilateral	Median
MAE	110	117	111	117
PSNR	33	33	31	33

Table 4 MAE and PSNR for Edge detection

Original Image	Sobel	Laplacian	prewitt
MAE	112	62	112
PSNR	3.36	10.11	3.36

Table 5 MAE and PSNR for Segmentation techniques

Original Image	Binary Otsu	K Means	Watershed
MAE	119	120	66
PSNR	2.15	2.02	10.65

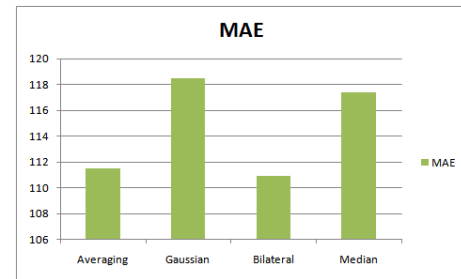


Fig. 8. MAE comparison plot for filters

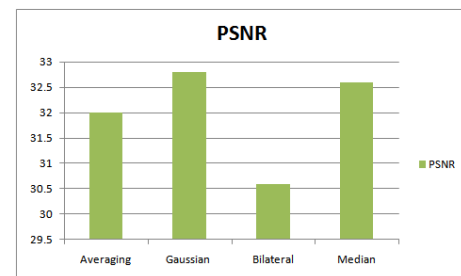


Fig. 9. PSNR comparison plot for filters

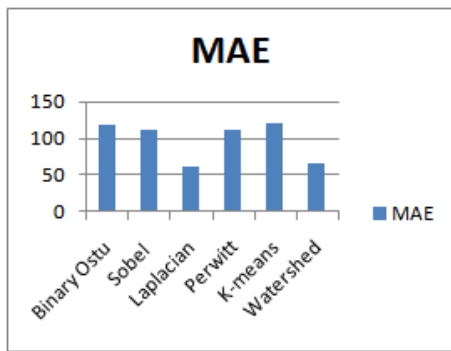


Fig. 10. MAE comparison plot for segmentation

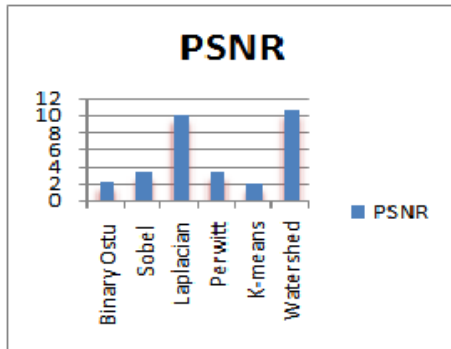


Fig. 11. PSNR comparison segmentation

VI. CONCLUSION

Leukemia is blood-forming cancer where abnormal cells grow rapidly in turn reducing normal cells. segmentation of cell of the nucleus is still challenging, here different filters, edge detection and segmentation techniques were applied from which Averaging filter, Laplacian edge detection and watershed segmentation were giving better results for ALL IDB2 images. with MAE 110,62 and 66 and PSNR 33, 2.15 and 10.65 respectively

REFERENCES

- [1] S. Mohapatra and D. Patra, "Automated cell nucleus segmentation and acute leukemia detection in blood microscopic images," 2010 International Conference on Systems in Medicine and Biology, 2010, pp. 49-54, doi: 10.1109/ICSMB.2010.5735344.
- [2] R. P and S. D. P, "Detection of Blood Cancer-Leukemia using K-means Algorithm," 2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS), 2021, pp. 838-842, doi: 10.1109/ICICCS51141.2021.9432244.
- [3] Nilkanth Mukund Deshpande, Shilpa Gite, Biswajeet Pradhan, Ketan Kotecha, Abdullah Alamri. Improved Otsu and Kapur approach for white blood cells segmentation based on LebTLBO optimization for the detection of Leukemia[J]. Mathematical Biosciences and Engineering, 2022, 19(2): 1970-2001. doi: 10.3934/mbe.2022093
- [4] F. Soni, L. Sahu, M. E. Getnet and B. Y. Reta, "Supervised Method for Acute Lymphoblastic Leukemia Segmentation and Classification Using Image Processing," 2018 2nd International Conference on Trends in Electronics and Informatics (ICOEI), 2018, pp. 1075-1079, doi: 10.1109/ICOEI.2018.8553937.
- [5] Toh, Leow Bin, et al. "Image segmentation for acute leukemia cells using color thresholding and median filter." Journal of Telecommunication, Electronic and Computer Engineering (JTEC) 10.1-5 (2018): 69-74.
- [6] H. N. Lim, M. Y. Mashor, and R. Hassan, "White blood cell segmentation for acute leukemia bone marrow images," Biomedical Engineering (ICoBE), 2012 International Conference on. pp. 357-361, 2012
- [7] A.P. Patil, "A Study of Segmentation Techniques to Detect Leukaemia in Microscopic Blood Smear Images," 2020 International Conference on Communication, Computing and Industry 4.0 (C2I4), 2020, pp. 1-6, doi: 10.1109/C2I451079.2020.9368928
- [8] Reena M.Roy, Ameer P.M., Segmentation of leukocyte by semantic segmentation model: A deep learning approach, Biomedical Signal Processing and Control, Volume 65, 2021, 102385, ISSN 1746-8094, <https://doi.org/10.1016/j.bspc.2020.102385>.
- [9] Aiswariya, P., and S. Manimekalai. "Global and Local Entropy Based Segmentation Model for Detecting Leukemia in Blood Images." Annals of the Romanian Society for Cell Biology (2021): 1914-1926.
- [10] D. R. Putri, A. Jamal and A. A. Septiandri, "Acute Lymphoblastic Leukemia Classification in Nucleus Microscopic Images using Convolutional Neural Networks and Transfer Learning," 2021 2nd International Conference on Artificial Intelligence and Data Sciences (AiDAS), 2021, pp. 1-6, doi: 10.1109/AiDAS53897.2021.9574176
- [11] N. H. A. Halim, M. Y. Mashor, A. S. Abdul Nasir, N. R. Mokhtar and H. Rosline, "Nucleus segmentation technique for acute Leukemia," 2011 IEEE 7th International Colloquium on Signal Processing and its Applications, 2011, pp. 192-197, doi: 10.1109/CSPA.2011.5759871.
- [12] D. Umamaheswari and S. Geetha, "Segmentation and Classification of Acute Lymphoblastic Leukemia Cells Toolled with Digital Image Processing and ML Techniques," 2018 Second International Conference on Intelligent Computing and Control Systems (ICICCS), 2018, pp. 1336-1341, doi: 10.1109/ICCONS.2018.8662950
- [13] J. Laosai and K. Chamnongthai, "Deep-Learning-Based Acute Leukemia Classification Using Imaging Flow Cytometry and Morphology," 2018 International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS), 2018, pp. 427-430, doi: 10.1109/ISPACS.2018.8923175.
- [14] F. Soni, L. Sahu, M. E. Getnet and B. Y. Reta, "Supervised Method for Acute Lymphoblastic Leukemia Segmentation and Classification Using Image Processing," 2018 2nd International Conference on Trends in Electronics and Informatics (ICOEI), 2018, pp. 1075-1079, doi: 10.1109/ICOEI.2018.8553937.
- [15] H. Nor Hazlyna et al., "Comparison of acute leukemia Image segmentation using HSI and RGB color space," 10th International Conference on Information Science, Signal Processing and their Applications (ISSPA 2010), 2010, pp. 749-752, doi: 10.1109/ISSPA.2010.5605410
- [16] M. Fatma and J. Sharma, "Identification and classification of acute leukemia using neural network," 2014 International Conference on Medical Imaging, m-Health and Emerging Communication Systems (MedCom), 2014, pp. 142-145, doi: 10.1109/MedCom.2014.7005992.
- [17] P. Kumar and S. M. Udawadia, "Automatic detection of acute myeloid leukemia from microscopic blood smear image," 2017 International Conference on Advances in Computing, Communications and Informatics (ICACCI), 2017, pp. 1803-1807, doi: 10.1109/ICACCI.2017.8126106.
- [18] Tina Babu, Tripty Singh and Deepa Gupta, "Colon cancer prediction using 2DReCA segmentation and hybrid features on histopathology images", in IET Image Processing, Vol. 14, No. 16, pp. 4144 - 4157, 2020, doi: 10.1049/iet-ipr.2019.1717. (Impact Factor - 1.995, Scopus Indexed, SCI Indexed)
- [19] Jithy Varghese and Tripty Singh, Venkatraman Bhat, and Moni Kuriakose. "Segmentation and Three Dimensional Visualization of Mandible Using Active Contour and Visualization Toolkit in Craniofacial Computed Tomography Images." Journal of Computational and Theoretical Nanoscience, Vol. 17, No. 1, pp. 61-67, 2020 (Scopus Indexed)
- [20] T. Babu, D. Gupta, T. Singh, and S. Hameed, "Prediction of normal & grades of cancer on colon biopsy images at different magnifications using minimal robust texture & morphological features," Indian Journal of Public Health Research & Development, 11(1), 695-701. 2020. doi: 10.37506/v11i1/2020/ijphrd/193905.
- [21] <https://homes.di.unimi.it/scotti/all/>